Flowsource - Database Backup and Restore

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1. Overview

The purpose of the document is to provide guidance on Database backup and Restore options for Flowsource installation on AWS, Azure and GCP Cloud environments. The details are based on the documents from the various Cloud providers and further information on these aspects can be found from the relevant Cloud Provider documentation.

2. AWS – RDS for PostgreSQL

Flowsource installation on EKS makes use of RDS for PostgreSQL. AWS provides the following options for Backup of AWS RDS For PostgreSQL:

- Automated Backups
- AWS Backup service
- Manual Backups with the help of Database Snapshots

The backup provide Point In Time Recovery (PITR) and the ability to restore the Database to a previous snapshot (taken once a day) or a particular point in time. The backups can be replicated to another region for DR scenarios.

https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/CHAP_CommonTasks.BackupRestore.html provides further details on Backup and Restore options for RDS

2.1 Automated Backups

Automated backups occur daily during the preferred backup window. This is automatically enabled when setting up the Database using the provided Terraform scripts if the backup retention period (value > 0) is provided. The backup window cannot overlap with the weekly maintenance window for the DB instance. If a backup window is not specified AWS automatically selects a 30 minute backup window.

The backup retention period can be specified between 0 to 35 days. Setting the value to 0 disables automated backups. If you want automated backup to be available for a longer period then you have to copy it to create a manual snapshot, which is retained until you delete it (https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/USER_CopySnapshot.html). Copy can be taken through AWS console or AWS CLI. .

If automated backup is not enabled this can be enabled anytime through the console or CLI.

For additional DR capability, you can configure Amazon RDS instance to replicate snapshots and transaction logs to a (paired) destination AWS region. When this is configured, RDS initiates copy of all snapshots and transaction logs as soon as they are ready on the DB instance. Replication of automated backups to another region can be enabled form the AWS console or CLI. In the source region replicated backups are listed under Current Region tab of Automated backup page. In destination region replicated backups are listed on the Replicated backup tabs of the Automated backup page. More information on this can be found at https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/USER_ReplicateBackups.html#:~:text=To%20restore%20a%20DB%20instance,navigation%20pane%2C%20choose%20Automated%20backups

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Automated backups can be viewed through AWS CLI or on the AWS console for RDS under Automated Backups menu.

2.1.1 Restoring Database from Automated Backup

This section describes how to restore a database instance. More details can be found from AWS documentation at https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/USER_RestoreFromSnapshot.html

You can restore a database from a snapshot or perform Point In Time recovery (PITR) as well to restore the DB instance to a specific point in time. The details of the earliest and latest restorable time can be viewed through the AWS console or AWS CLI. The earliest restorable time depends on the backup retention period.

The Database can be restored from Console (under RDS) or through CLI as well. If backup has been replicated to another region, then you can restore from replicated backup as well either through console or CLI.

RestoreDBInstanceFromDBSnapshot permission is needed to restore DB from a snapshot. In order to perform PITR RestoreDBInstanceToPointInTime permission is needed.

2.2 AWS Backup Service

AWS is a fully managed backup service that makes it easy to centralize and automate backup across AWS services. You can manage backup of RDS databases in AWS Backup as well. Details on the same can be found at https://docs.aws.amazon.com/aws-backup/latest/devguide/whatisbackup.html

Backups managed by AWS Backup can be viewed through the RDS Console by navigating to Snapshots -> Backup Service tab

In order to backup RDS using AWS Backup service, a backup plan needs to be created. Backups are stored and organized in a Vault. For RDS AWS Backup supports continuous backup and PITR in addition to snapshot backups.

When using AWS Backup service for backing up RDS, as per RDS terminology this is considered as manual backup (similar to manual snapshots).

More details specific to AWS Backup for RDS can be found at https://docs.aws.amazon.com/aws-backup/latest/devguide/rds-backup.html

AWS Backup service can be used in conjunction with AWS RDS Automated backup. AWS Backup checks if you have previously used Amazon RDS to create an automated backup. If an automated backup exists, AWS Backup creates a copy of this snapshot (copy-db-snapshot operation). If no extant backup exists, AWS Backup creates a snapshot of the instance you indicate, instead of a copy (create-db-snapshot operation). However in this case it is important to ensure that the AWS backup plan window and the Amazon RDS window do not coincide else it could result in conflicting backups.

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2.2.1 Restoring Database from AWS Backup

Once the RDS has been backed up at least once through AWS Backup service, it is considered a protected resource. The database can be restored through the console or CLI. Refer https://learn.microsoft.com/en-us/azure/postgresql/flexible-server/concepts-backup-restore for additional details

3. Azure – Azure Database for PostgreSQL Flexible Server

Flowsource installation on AKS makes use of Azure Database for PostgreSQL – Flexible Server. Azure provides the following options for backup and restore of Azure Database for PostgreSQL – Flexible Server

- Automated Backups
- Azure Backup Service – Please note that this is in Preview for Azure Database for PostgreSQL – Flexible Server

The backup provide Point In Time Recovery (PITR) and the ability to restore the Database.

https://learn.microsoft.com/en-us/azure/postgresql/flexible-server/concepts-backup-restore provides further details on Backup and Restore options for Azure Database for PostgreSQL Flexible Server.

3.1 Automated Backups

Azure Database for PostgreSQL flexible server automatically performs regular backups of your server. Azure manages the backup window, and you can't customize the same. The first full snapshot is scheduled immediately after server creation. Subsequent snapshots are incremental and occur once a day.

Azure Database for PostgreSQL flexible server provides the following options for backups redundancy:

- Zone redundant backup storage – The option is automatically chosen for regions that support availability zones. Multiple copies are not only stored within the same AZ but also replicated to other AZ within the region.
- Locally redundant backup storage – The option is automatically chosen for regions that don't support availability zones. Multiple copies of backup are stored within the same datacentre.
- Geo-redundant backup storage – This option can only be chosen at server creation. After server is provisioned, you can't change the backup storage redundancy option. With this option in addition to three copies of backup stored within the region the server is hosted, data is replicated to a geo-paired region. Refer https://learn.microsoft.com/en-us/azure/reliability/cross-region-replication-azure for the Azure paired regions.
  This option allows you to restore your server in a different region in the event of a disaster and thus can be used as a DR strategy. Please note however that you can

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restore only upto the last available backup data that is available at the paired region. Currently PITR of geo-redundant backups is not supported.

In addition to regular backup snapshots, Azure Database for PostgreSQL Flexible Server also backs up transaction logs when the write-ahead log (WAL) file is ready to be archived. You can use these backups to restore a server to any point in time (PITR) within your configured backup retention period. Transaction log backups happen at varied frequencies, depending on the workload and when the WAL file is filled and ready to be archived. In general, the delay (recovery point objective, or RPO) can be up to 15 minutes.

Backups are retained based on the retention period that you set for the server. You can select a retention period between 7 (default) and 35 days. You can change the retention period post server creation as well.

Unlike AWS, in Azure the automated backups cannot be exported or copied for longer term retention. If longer term retention is required then the following options are available:

- Manual backup using PostgreSQL tool pg_dump.
  For examples see https://learn.microsoft.com/en-us/azure/postgresql/migrate/how-to-migrate-using-dump-and-restore?tabs=psql
- Use Azure Backup Service (which is in Preview)

Note: If you delete a server, all backups that belong to the server are retained for five days in the service post which they are deleted and can't be recovered. You can follow the steps at https://learn.microsoft.com/en-us/azure/postgresql/flexible-server/how-to-restore-dropped-server to restore the deleted server, however it will work only if the backup for the server is still available and not deleted. Do also note that the restoration process is not always guaranteed.

To help protect server resources from accidental deletion by using a “delete lock” which is a type of management lock in Azure.

3.1.1 Restoring Database from Automated Backup

With Automated backup and continuous backup of Transaction logs, you can restore the Database as per the following options:

- Latest Restore Point
- Custom Restore Point – This option allows you to choose any point in time within the retention period
- Fast Restore Point – This option allows you to restore the server in the fastest time possible within the retention period. It restores the database to the last database snapshot and does not apply the recovery logs

The time to restore using the latest and custom restore points varies based on factors such as volume of transactions processed since last backup and number of databases being recovered simultaneously in the same region.

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In Azure Database for PostgreSQL flexible server, performing a restore creates a new Server. If you configure your server within a virtual network, you can restore to the same virtual network or to a different virtual network. However, you can't restore to public access. Similarly, if you configured your server with public access, you can't restore to private virtual network access.

If geo-redundant backup is enabled, then you can restore it to a geo-paired region. In this case you can restore only till the last available backup data that is available at the paired region. PITR of geo-redundant backup is currently not supported.

Do also note that the server is restored as a single-instance Azure Database for PostgeSQL flexible server instance (even if the source server was HA). After restore is complete you can optionally configure the server with high availability.

Please refer https://learn.microsoft.com/en-us/azure/postgresql/flexible-server/concepts-backup-restore for more details on Restore from Automated backups.

The restore operation can be carried out either from the Azure Portal or CLI. Refer the guides as below for the steps:

- Azure Portal - https://learn.microsoft.com/en-us/azure/postgresql/flexible-server/how-to-restore-server-portal
- Azure CLI - https://learn.microsoft.com/en-us/azure/postgresql/flexible-server/how-to-restore-server-cli

3.2 Azure Backup Service

Backing up of Azure Database for PostgreSQL flexible server using backup service is currently under preview. This allows backups to be retained for upto 10 years. Refer https://learn.microsoft.com/en-us/azure/backup/backup-azure-database-postgresql-flex-overview for more details.

The backups are copied to an isolated storage environment outside of customer tenant and subscription. Refer https://learn.microsoft.com/en-us/azure/backup/backup-azure-database-postgresql-flex for configuring backup service for Azure Database for PostgreSQL flexible server from the Azure Portal.

3.2.1 Restoring Database from Azure Backup Service

Backup data from Azure Backup service is stored in the Backup Vault as a blob within the Microsoft tenant. Before restoring from Azure Database for PostgreSQL Flexible server backups, the required permissions need to be ensured. Follow the steps specified at https://learn.microsoft.com/en-us/azure/backup/restore-azure-database-postgresql-flex in order to restore Azure PostgreSQL-Flexible database. During restore backup data is copied from one storage account to another across tenants. Once the backup data is restored into the storage account, you can use the pg_restore utility to restore the database.

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4. GCP – Alloy DB

Flowsource installation on GCP makes use of AlloyDB. GCP provides the following options for backup of AlloyDB servers

- Continuous Backup and Recovery
- Discrete backups - These can be on-demand or enabled as automated backup schedule to create regular backupsx.

Please refer https://cloud.google.com/alloydb/docs/backup/overview for further details on backup and restore of AlloyDB.

Refer the below mentioned links for further details on managing backups including listing and viewing details of the same

https://cloud.google.com/alloydb/docs/backup/list

https://cloud.google.com/alloydb/docs/backup/view

https://cloud.google.com/alloydb/docs/backup/delete

4.1 Continuous Backup

Continuous backup is enabled by default on all AlloyDB clusters unless explicitly disabled. It lets you restore cluster in the same region (can be different project) to any moment of its recent history with microsecond granularity. By default the retention days for continuous backups is 14 days. The same can however be configured to a minimum of 1 day and maximum 35 days either from the Cloud Console or CLI.

Continuous Backup allows you to restore your cluster with a near zero RPO i.e. you can restore your cluster to a state that it had moments before a catastrophic event. Refer https://cloud.google.com/alloydb/docs/backup/configure for details on configuring Continuous backup from the Cloud Console or CLI. You can also disable and enable continuous backup at any point of time either through the console or CLI. Point in time recovery (PITR) becomes unavailable for cluster if continuous backup is disabled. It becomes available again only if you re-enable continuous backup. The recovery window begins afresh with the first subsequent backup that Alloy DB creates. After configuring it might take upto 24 hours for continuous backup and recovery to take a backup and enable its continuous recovery capability. If you want to accelerate the backup operation without waiting for 24 hours and to avoid data loss, then you can take an on-demand backup after you configure the plan.

AlloyDB allows you to delete continuous backups based on certain conditions:

- You have disabled continuous backup and recovery on this cluster.
- The backup that you want to delete is older than the cluster's configured recovery window for continuous backup and recovery.

Additionally it doesn't let you delete backups that has other backups based on it. These backups are incremental backups and rely on existence of specific previous backups.

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4.1.1 Restoring from Alloy DB Continuous Backup

Alloy DB lets you restore fully a cluster to a specific point in time within the continuous backup retention/recovery period. You can only perform PITR after Alloy DB has taken one new backup or a subsequent backup post disabling and enabling continuous backup.

When you restore from continuous backup you create a new cluster in the same region (can be different project). You can use either the CLI or the Console to restore the cluster. AlloyDB creates a new cluster and initiates a long running operation to load the backed up data into the cluster's storage. Once the cluster is created you need to create the primary instance for it. That instance lets you access the restored data. The new instance's configuration need not exactly match the source cluster. You can optionally also create read-replicas for the new cluster. Unlike restoring from a backup, PITR requires the original cluster to still exist. Thus you cannot perform PITR on a deleted cluster.

The new cluster is completely independent of the original cluster including having its own backup configurations.

4.2 On-demand backups

You can create an on-demand backup at any point in time using the Google Cloud CLI or console. On-Demand backups are full backups with each such backup holding all the data that was in its cluster databases when the backup operation began.

On-demand Alloy DB backup allows you to create a backup in the same region as the original cluster (default) or a cross region backup. If you don't specify a location backups are stored in the original location of the Alloy DB cluster.

Cross-region backup option is not available through the Cloud Console and can be performed only through the CLI (version 484.0.0 or higher). Cross-region backup is useful to restore cluster in another region.

On-demand backups can have a retention period of upto 1 year. Backups older than the retention period might still appear when viewing the project's backups however they are expired and subject to automated deletion. If you no longer require an on-demand backup before the automated deletion you can manually delete it either using the console or through the CLI.

Refer https://cloud.google.com/alloydb/docs/backup/create-on-demand for details on configuring on-demand backup through Cloud Console and CLI.

4.2.1 Restoring from Alloy DB On-Demand Backup

You can restore an Alloy DB cluster from backup in the same region as that of the backup. When restoring from a backup Alloy DB creates a new cluster in the region of the backup (this is not the same as region of original cluster since in case of on-demand backup, the backup could be stored in a cross region location). The backup used to restore can be in the same project or in a different project. Once the cluster is created you need to create the primary instance for it. That instance lets you access the restored data. The new instance's

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configuration need not exactly match the source cluster. You can optionally also create read-replicas for the new cluster. The new cluster is completely independent of the original cluster including having its own backup configurations.

Point in time recovery is not possible by using on-demand backup.

You can use this method to restore a cluster to a state which is older than the recovery window or to restore a cluster that is no longer online.

4.3 Automated backups

You can enable automated backups schedule in AlloyDB so that Alloy DB creates additional backups regularly according to the defined schedule. Automated backups are incremental, similar to continuous backups. These can only be created in the same region as the original cluster (unlike on-demand backups).

Automated backups have a retention period of upto 1 year. Default retention period is 14 days. Backups older than the retention period might still appear when viewing the project's backups however they are expired and subject to automated deletion.

Each Alloy DB cluster can have one customizable backup configuration which includes:

- Schedule for taking backups. You can specify days of the week and specific times on those days
- A retention policy expressed either as number of days or number of backups
- The amount of time Alloy DB waits before cancelling an incomplete attempt to take a new automated backup. AlloyDB uses a default time of 1 hour.
- A data encryption preference

Please note that the Google Cloud console only allows you to enable/disable automated backup schedule which will configure the backup schedule according to default configuration (or previously configured values). In order to specify configurations you need to sue the CLI. Each time the CLI is used to configure an automated backup schedule it replaces the cluster's current configuration.

Refer https://cloud.google.com/alloydb/docs/backup/configure for configuring automated backup through the Console and CLI.

4.3.1 Restoring from Alloy DB Automated Backup

You can restore an Alloy DB cluster from backup in the same region as that of the original cluster. When restoring from a backup Alloy DB creates a new cluster in the region of the backup (or in this case the original cluster region). The backup used to restore can be in the same project or in a different project. Once the cluster is created you need to create the primary instance for it. That instance lets you access the restored data. The new instance's configuration need not exactly match the source cluster. You can optionally also create read-replicas for the new cluster. The new cluster is completely independent of the original cluster including having its own backup configurations.

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Point in time recovery is not possible by using automated backup. You can use this method to restore a cluster to a state which is older than the recovery window or to restore a cluster that is no longer online.

4.4 Backup & Restore Features Summary across AWS, GCP and Azure

The table below provides a comparison on the backup & restore features across AWS, GCP and Azure for the respective databases.

Feature	AWS	GCP	Azure
Automated Backup (Same Region)	Yes (If backup retention > 0)	Yes (Unless disabled)	Yes
Automated Backup (Cross Region)	Yes (to paired region) – By enabling backup replication	No	If Geo-Redundant storage is configured
Backup schedule configuration	User specified. Cannot overlap maintenance window	User specified configuration if automated schedule enabled and configured	Azure managed schedule. Taken once daily. User cannot specify schedule.
On Demand Backup/Snapshots	Yes (Same Region as well as Cross Region)	Yes (Same Region as well as Cross Region)	No
PITR (Same Region)	Yes	Yes	Yes
PITR (Cross Region)	Yes	No	No
PITR Granularity (RPO)	Upto 5 minutes	Near 0 (in microseconds)	Upto 5 minutes
PITR – Earliest Restorable Time	Upto 35 days	Upto 35 days	Upto 35 days
Backup Retention	Automated backups - 0 to 35 days (Default 7 days) Manual Snapshots.- Until deleted	Continuous Backups – 1 to 35 days (Default 14 days) Automated Backups – 1 day to 1 year (Default 14 days) On Demand Backup – 1 year	7 days to 35 days
Ability to export/copy backups	Yes	No	No
Long Term Retention Options	Copy Backups periodically Use AWS Backup Service	Max retention is 1 year. Other native Postgres utilities such as pg_dump can be explored for longer retention	Max retention is 35 days. Use native pg_dump utility for longer term requirements.

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			Azure Backup Service can provide retention upto 10 years but is currently under preview
Recovery of deleted cluster	Yes – From Final Snapshot (if taken before deletion) or from other manual snapshots. PITR possible if automated backups retained (upto the automated backup retention period)	Yes – From Automated or On-Demand backups. PITR not possible	Upto 5 days post deletion recovery can be attempted through documented process – though not guaranteed
Central Backup Service and Vault Available	Yes – AWS Backup	No GCP Backup service does not support Alloy DB	No Azure Backup service support for Azure Database for PostgreSQL Flexible Server is currently under preview

5. AWS – Redshift Serverless

AWS Redshift Cluster is used for Devops Insights metrics. AWS provides the following options for backup of Redshift Serverless Clusters

- Manually created snapshots
- Recovery Points automatically created by Redshift

More details on the backup and restore options for AWS Redshift Serverless cluster can be found at https://docs.aws.amazon.com/redshift/latest/mgmt/serverless-snapshots-recovery-points.html

5.1 Manually created snapshots

A backup in Amazon Redshift Serverless is a point-in-time representation of the objects and data in your namespace. Manual snapshots allow you to restore a snapshot to a serverless namespace or to a provisioned cluster.

Snapshots can be taken from the AWS Redshift Serverless console or through the CLI. The default retention period for a manual snapshot is Indefinite (in which case snapshot needs to

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be manually deleted when not required). You can optionally chose a value between 1-3653 days. You can also update the retention period of manual snapshot at a later point in time.

You can also configure Redshift Serverless to automatically copy snapshot to another AWS region. When you create a snapshot in the *source* AWS Region, it's copied to a *destination* Region. When configuring copying manual snapshots you can also specify the retention period. The retention period in the destination region is different from the source region. The snapshot in the destination region can be used to restore data to a Redshift Serverless namespace or provisioned cluster in the destination region.

To precisely control when to take a snapshot, you can create a snapshot schedule for specific namespaces. This can be done only through API and CLI.

5.1.1 Restoring from Redshift Serverless Manual Snapshots

Restoring a snapshot to a serverless namespace replaces the current database with the database in the snapshot. You can restore a manual serverless snapshot to a serverless namespace or provisioned cluster.

Restoring a snapshot to a serverless namespace takes place in two phases. The first phase completes in a few minutes in which it restores the data to your namespace and makes it available for queries. The second phase of restoration is where the database is tuned which can cause minor performance issues. The second phase can last from a few hours to several days and performance progressively improves as the database gets tuned.

You need an existing serverless namespace for restoring snapshot to serverless namespace. When restoring to a provisioned cluster, a cluster is created based on the configuration provided.

You can also restore a specific table from a manual snapshot. When you do so you specify the source snapshot, database, schema, table and the target database, schema and new table name. The new table name can’t have the same name as existing table. If you want to replace an existing table by restoring a table, you must first rename or drop the table before you restore the table. The target table is created with the source tables attributes, column definitions and column attributes except for foreign keys. Also any dependencies such as views or permissions granted on the source table aren’t applied to the target tables and needs to be done separately. You can restore only one table at a time.

5.2 Recovery Points

A backup in Amazon Redshift Serverless is a point-in-time representation of the objects and data in your namespace. Recovery points are created every 30 minutes and saved for 24 hours. You can convert a recovery point to a snapshot for longer retention. Recovery points can be restored to a serverless namespace or to a provisioned cluster.

You cannot change the retention period for a recovery point.

You can also configure Redshift Serverless to automatically copy recovery points to another AWS region. When a recovery point is created in the *source* AWS Region, it's copied to

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a *destination* Region. The retention period in the destination region will also be 1 day and cannot be changed. Once the recovery point is copied to the destination region, you can use it restore data to a Redshift Serverless namespace or provisioned cluster in the destination region.

5.2.1 Restoring from Redshift Serverless Recovery Points

You can restore a serverless recovery point to a serverless namespace or provisioned cluster.

You need an existing serverless namespace for restoring snapshot to serverless namespace and its status should be available. When restoring to a provisioned cluster, a cluster is created based on the configuration provided.

You can also restore a specific table from a recovery point. When you do so you specify the source snapshot, database, schema, table and the target database, schema and new table name. The new table name can't have the same name as existing table. If you want to replace an existing table by restoring a table, you must first rename or drop the table before you restore the table. The target table is created with the source tables attributes, column definitions and column attributes except for foreign keys. Also any dependencies such as views or permissions granted on the source table aren't applied to the target tables and needs to be done separately. You can restore only one table at a time.

6. AWS – Redshift Provisioned Cluster

AWS Redshift Cluster is used for Devops Insights metrics. AWS provides the following options for backup of Redshift Provisioned Clusters

- Manual snapshots
- Automated Snapshots
- AWS Backup Service

More details on the backup and restore options for AWS Redshift Serverless cluster can be found at https://docs.aws.amazon.com/redshift/latest/mgmt/working-with-snapshots.html

6.1 Automated Snapshots

Snapshots are point-in-time backups. Amazon Redshift automatically takes incremental snapshots and these retain all of the data required to restore a cluster from a snapshot. To ensure that your backups are always available to your cluster, Amazon Redshift stores snapshots in an internally managed Amazon S3 bucket that is managed by Amazon Redshift. Automated snapshots are enabled by default when you create a cluster.

When Automates snapshots are enabled for a cluster Amazon Redshift takes a snapshot about every 8 hours, or following every 5GB of data changes or whichever comes first. The shortest time between automated snapshot creation is 15 minutes. Alternatively you can create a snapshot schedule to control when automated snapshots are taken. If you're using custom schedules, the minimum amount of time between automated snapshots is one hour.

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A *snapshot schedule* is a set of schedule rules. You can define a simple schedule rule based on a specified interval, such as every 8 hours or every 12 hours. You can also add rules to take snapshots on certain days of the week, at specific times, or during specific periods.

The default retention period for automated snapshot is 1 day, but you can modify it from 0 days to upto 35 days. The automated snapshots are deleted at the end of retention period. To disable automated snapshots set the retention period to 0. If you disable automated snapshots Amazon redshift stops taking automated snapshots and deletes any existing automated snapshots for the cluster.

Automated snapshots cannot be deleted manually. If you want to keep automated snapshots for a longer period you can create a copy of it as a manual snapshot.

You can also configure Amazon Redshift to automatically copy snapshots to another AWS region. When you enable Amazon Redshift to automatically copy snapshots to another AWS region, you specify the destination AWS region and the retention period to keep them in the destination region. The retention period in the destination region is different from the source region. The snapshot in the destination region can be used to restore data to a Redshift Serverless namespace or provisioned cluster in the destination region.

6.1.1 Restoring from Redshift Provisioned Cluster Automated Snapshots

A snapshot contains data from any databases that are running on your cluster. It also contains information about your cluster, including the number of nodes, node type, and admin user name. If you restore your cluster from a snapshot, Amazon Redshift uses the cluster information to create a new cluster. Then it restores all the databases from the snapshot data. For the new cluster created from the original snapshot, you can choose the configuration, such as node type and number of nodes. The cluster is restored in the same AWS Region and a random, system-chosen Availability Zone, unless you specify another Availability Zone in your request.

When you restore from a snapshot, Amazon Redshift creates a new cluster and makes the new cluster available before all of the data is loaded, so you can begin querying the new cluster immediately. The cluster streams data on demand from the snapshot in response to active queries, then loads the remaining data in the background.

You can't use a snapshot to revert an active cluster to a previous state.

You can restore an automated snapshot from Redshift Provisioned cluster to a serverless namespace as well. Refer https://docs.aws.amazon.com/redshift/latest/mgmt/managing-snapshots-console.html#snapshot-restore-provisioned-to-serverless for more details. Restoring a serverless namespace from a snapshot replaces all of the namespace's databases with databases in the snapshot.

You can restore a single table from a snapshot instead of restoring an entire cluster. When you restore a single table from a snapshot, you specify the source snapshot, database, schema, and table name, and the target database, schema, and a new table name for the restored table. The new table name can't be the name of an existing table. If you want to replace an existing table by restoring a table, you must first rename or drop the table before you restore

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the table. The target table is created with the source tables attributes, column definitions and column attributes except for foreign keys. Also any dependencies such as views or permissions granted on the source table aren't applied to the target tables and needs to be done separately. You can restore only one table at a time.

6.2 Manual Snapshots

Snapshots are point-in-time backups. To ensure that your backups are always available to your cluster, Amazon Redshift stores snapshots in an internally managed Amazon S3 bucket that is managed by Amazon Redshift.

You can take a manual snapshot any time. By default manual snapshots are retained indefinitely, even after you delete the cluster. You can specify the retention period when you create a manual snapshot, or you can change the retention period by modifying the snapshot.

Amazon Redshift has a quota that limits the total number of manual snapshots that you can create; this quota is per AWS account per AWS Region

You can also configure Amazon Redshift to automatically copy snapshots to another AWS region. When you enable Amazon Redshift to automatically copy snapshots to another AWS region, you specify the destination AWS region and the retention period to keep them in the destination region. The retention period in the destination region is different from the source region. The snapshot in the destination region can be used to restore data to a Redshift Serverless namespace or provisioned cluster in the destination region.

6.2.1 Restoring from Redshift Provisioned Cluster Manual Snapshots

A snapshot contains data from any databases that are running on your cluster. It also contains information about your cluster, including the number of nodes, node type, and admin user name. If you restore your cluster from a snapshot, Amazon Redshift uses the cluster information to create a new cluster. Then it restores all the databases from the snapshot data. For the new cluster created from the original snapshot, you can choose the configuration, such as node type and number of nodes. The cluster is restored in the same AWS Region and a random, system-chosen Availability Zone, unless you specify another Availability Zone in your request.

When you restore from a snapshot, Amazon Redshift creates a new cluster and makes the new cluster available before all of the data is loaded, so you can begin querying the new cluster immediately. The cluster streams data on demand from the snapshot in response to active queries, then loads the remaining data in the background.

You can't use a snapshot to revert an active cluster to a previous state.

You can restore an automated snapshot from Redshift Provisioned cluster to a serverless namespace as well. Refer https://docs.aws.amazon.com/redshift/latest/mgmt/managing-snapshots-console.html#snapshot-restore-provisioned-to-serverless for more details. Restoring a serverless namespace from a snapshot replaces all of the namespace's databases with databases in the snapshot.

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You can restore a single table from a snapshot instead of restoring an entire cluster. When you restore a single table from a snapshot, you specify the source snapshot, database, schema, and table name, and the target database, schema, and a new table name for the restored table. The new table name can't be the name of an existing table. If you want to replace an existing table by restoring a table, you must first rename or drop the table before you restore the table. The target table is created with the source tables attributes, column definitions and column attributes except for foreign keys. Also any dependencies such as views or permissions granted on the source table aren't applied to the target tables and needs to be done separately. You can restore only one table at a time.

6.3 AWS Backup Service for Redshift Clusters

AWS Backup is a fully managed service that helps you centralize and automate data protection across AWS services. Using AWS Backup for Amazon Redshift, you can configure data protection policies and monitor activity for different Amazon Redshift resources in one place. You can also create and store snapshots on Amazon Redshift provisioned clusters. Amazon Redshift is natively integrated with AWS Backup. That lets you define your backup plans and assign Amazon Redshift resources to the backup plans. AWS Backup automates the creation of Amazon Redshift manual snapshots, and securely stores these snapshots in an encrypted backup vault that you designate in your backup plan.

Refer https://docs.aws.amazon.com/redshift/latest/mgmt/managing-aws-backup.html for more details.

AWS Backup doesn't support Amazon Redshift Serverless. You can't use AWS backup to manage Amazon Redshift automated backups.

To protect resources on your Amazon Redshift provisioned clusters, you can use the AWS Backup console, or programmatically use the AWS Backup API or AWS Command Line Interface (AWS CLI). When you need to recover a resource, you can use either the AWS Backup console or the AWS CLI to find and recover the resource you need.

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